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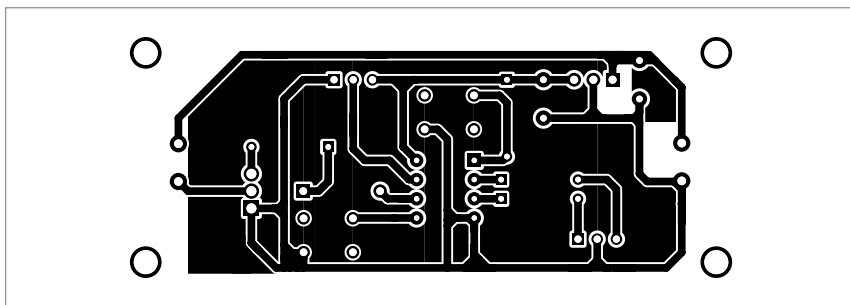


Fig. 3: Actual-size PCB layout for liquid dispenser

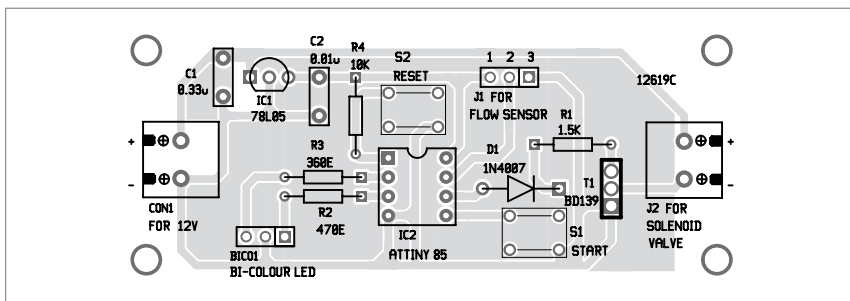


Fig. 4: Component layout of the PCB

Software code

The source code is developed using Arduino IDE. By default, Arduino IDE does not support ATtiny85 MCU. So, we need to add ATtiny board to Arduino IDE as follows:

Open File→Preferences and copy the below-mentioned link in the Additional Boards Manager URL: https://raw.githubusercontent.com/damellis/attiny/ide-1.6.x-boards-manager/package_damellis_attiny_index.json.

After this is done, open Tools→Board→Board Manager. After opening Boards Manager, scroll down the list and find 'attiny' by Davis A. Mellis. Click on that and install it. After installing it, the ATtiny will be listed in the Boards menu.

Now, select ATtiny25/45/85 under Tools→Board. Then select ATtiny85 under Tools→Processor and select 8MHz (internal) under Tools→Clock.

Next, make Arduino Uno an ISP programmer. For this, grab your Arduino Uno, connect it to your PC, and

open Arduino IDE. Make sure you have selected the Arduino/Genuino Uno as your board, and open the ArduinoISP sketch from File→Examples option. Upload the ArduinoISP sketch to your Arduino Uno. Your Arduino Uno is now an ISP programmer.

By default, the ATtiny85 runs at 1MHz. To make it to run at 8MHz, select Tools→Burn Bootloader option. Make sure that Arduino Uno board is programmed as ISP, and it is connected with the ATtiny85 through ISP pins as per Table 1. Upload the source code *Simple_Liquid_Dispenser.ino* by using the same Arduino board programmed as ISP.

Construction and testing

An actual-size PCB layout for the simple liquid dispenser is shown in Fig. 3 and its component layout in Fig. 4. After assembling the circuit on PCB, enclose it in a suitable plastic box. Install the flow sensor and solenoid valve according to your application. **EFY**

EFY Note
The source code of this project is available for free download at source.efymag.com



A. Samiuddin, a circuit designer, is B.Tech in electrical and electronics engineering. His interests include LED lighting, power electronics, microcontrollers, and Arduino programming